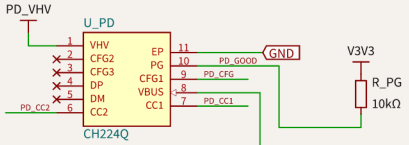
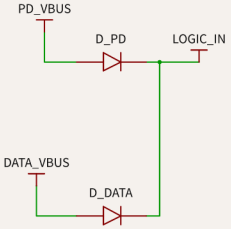
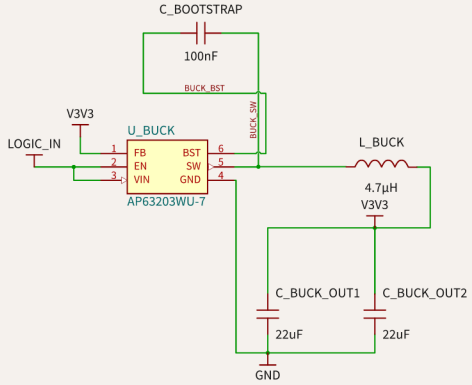
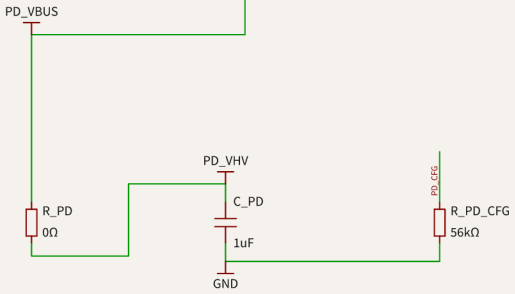

J_PD
PD_VBUS
EH2
EH1
EH4
EH3
SBU2
CC1
DN2
DP1
VBUS2
VBUS1
GND2
GND1
CC2
SBU1
DP2
DN1
GND
PD_CC1
PD_CC2
GND
TYPE-C-31-M-12

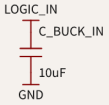

PD_VHV
U_PD
VHV
CFG2
CFG3
DP
DM
CC2
EP
PG
CFG1
VBUS
CC1
GND
PD_GOOD
PD_CFG
PD_CC1
PD_CC2
CH224Q
V3V3
R_PG
10kΩ

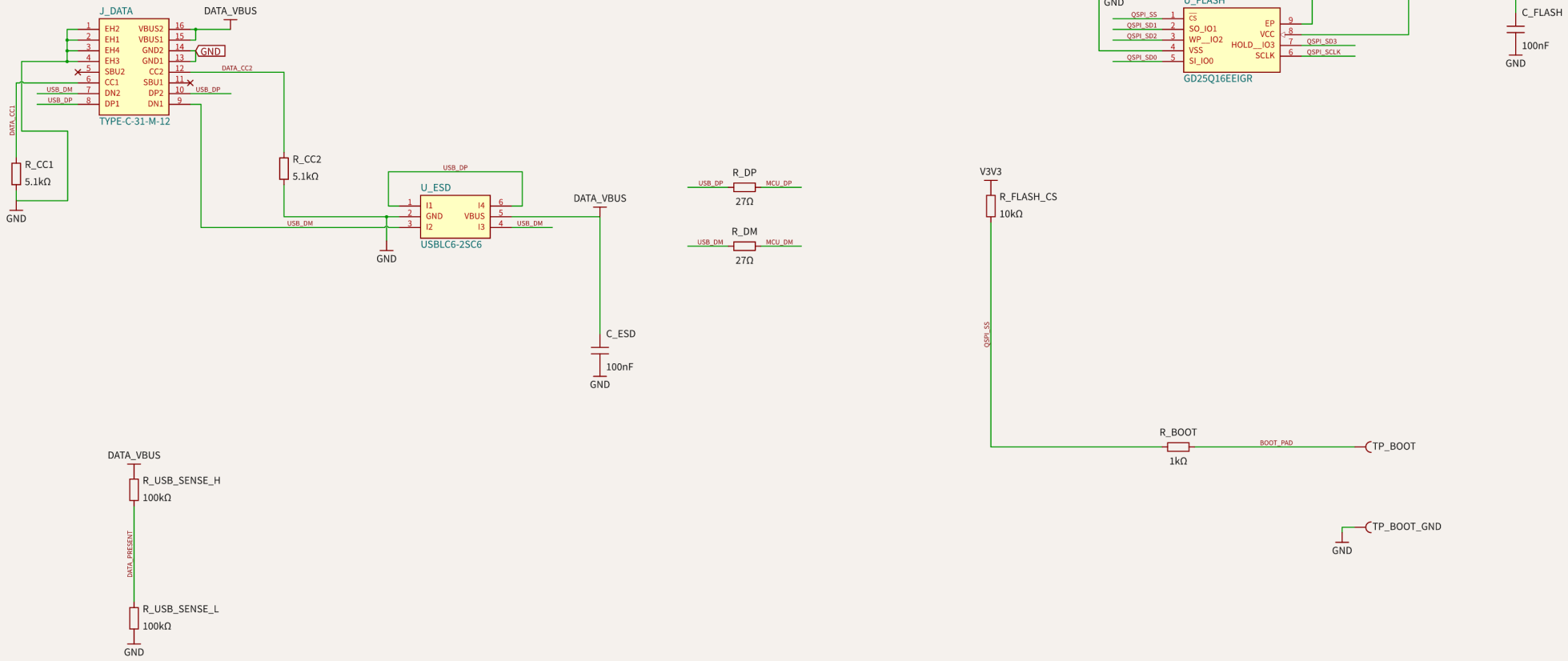

PD_VBUS
D_PD
LOGIC_IN
DATA_VBUS
D_DATA

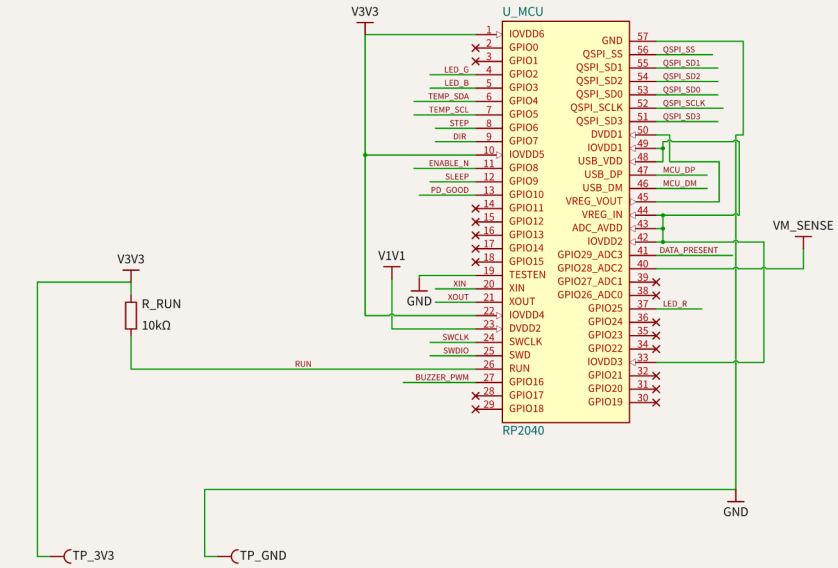

C_BOOTSTRAP
100nF
BUCK_BST
BUCK_SW
V3V3
LOGIC_IN
U_BUCK
FB
EN
VIN
BST
SW
GND
AP63203WU-7
L_BUCK
4.7µH
V3V3
C_BUCK_OUT1
22uF
C_BUCK_OUT2
22uF
GND

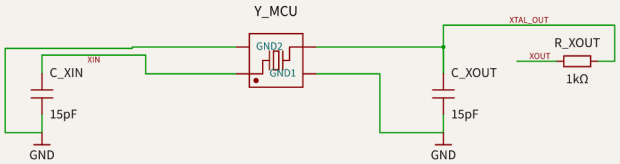

PD_VBUS
R_PD
0Ω
PD_VHV
C_PD
1uF
GND
PD_CFG
R_PD_CFG
56kΩ

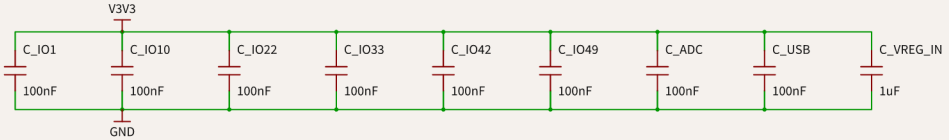

PD_VBUS
D_TVS
GND

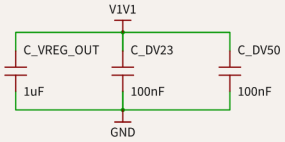

LOGIC_IN
C_BUCK_IN
10uF
GND

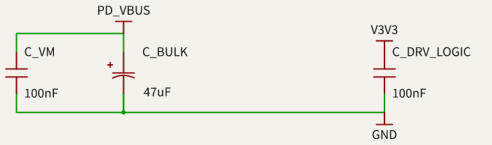

J_DATA
TYPE-C-31-M-12
DATA_VBUS
GND
DATA_CC2
USB_DM
USB_DP
DATA_CC1
R_CC1
5.1kΩ
R_CC2
5.1kΩ
U_ESD
USBLC6-2SC6
C_ESD
100nF
R_DP
27Ω
R_DM
27Ω
MCU_DP
MCU_DM
U_FLASH
GD25Q16EEIGR
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
V3V3
C_FLASH
100nF
R_FLASH_CS
10kΩ
R_BOOT
1kΩ
BOOT_PAD
TP_BOOT
TP_BOOT_GND
R_USB_SENSE_H
100kΩ
R_USB_SENSE_L
100kΩ
DATA_PRESENT

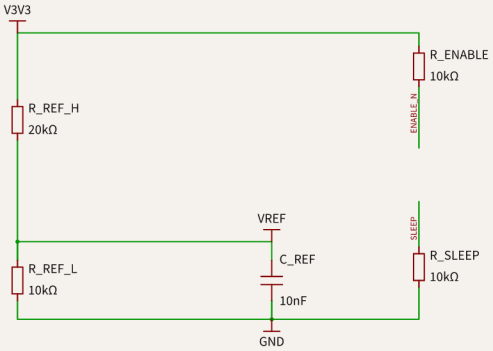

V3V3
U_MCU
IOVDD6
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
IOVDD5
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
TESTEN
XIN
XOUT
IOVDD4
DVDD2
SWCLK
SWD
RUN
GPIO16
GPIO17
GPIO18
GND
QSPI_SS
QSPI_SD1
QSPI_SD2
QSPI_SD0
QSPI_SCLK
QSPI_SD3
DVDD1
IOVDD1
USB_VDD
USB_DP
USB_DM
VREG_VOUT
VREG_IN
ADC_AVDD
IOVDD2
GPIO29_ADC3
GPIO28_ADC2
GPIO27_ADC1
GPIO26_ADC0
GPIO25
GPIO24
GPIO23
GPIO22
IOVDD3
GPIO21
GPIO20
GPIO19
LED_G
LED_B
TEMP_SDA
TEMP_SCL
STEP
DIR
ENABLE_N
SLEEP
PD_GOOD
MCU_DP
MCU_DM
DATA_PRESENT
LED_R
RP2040
V1V1
GND
VM_SENSE
R_RUN
10kΩ
RUN
TP_3V3
TP_GND
SWCLK
TP_SWCLK
SWDIO
TP_SWDIO
RUN
TP_RUN

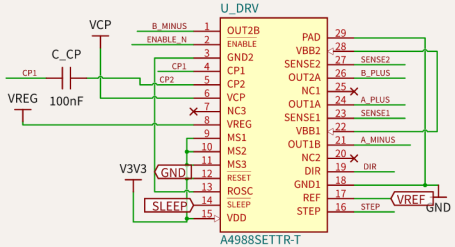

Y_MCU
GND2
GND1
XIN
C_XIN
15pF
GND
XTAL_OUT
R_XOUT
XOUT
1kΩ
C_XOUT
15pF
GND

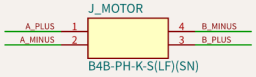

V3V3
C_IO1
C_IO10
C_IO22
C_IO33
C_IO42
C_IO49
C_ADC
C_USB
C_VREG_IN
100nF
100nF
100nF
100nF
100nF
100nF
100nF
100nF
1uF
GND

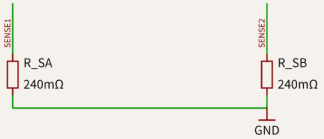

V1V1
C_VREG_OUT
C_DV23
C_DV50
1uF
100nF
100nF
GND

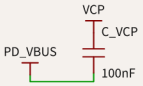

PD_VBUS
C_VM
100nF
C_BULK
47uF
V3V3
C_DRV_LOGIC
100nF
GND

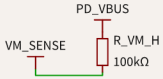

V3V3
R_REF_H
20kΩ
R_ENABLE
10kΩ
ENABLE_N
VREF
C_REF
10nF
SLEEP
R_SLEEP
10kΩ
R_REF_L
10kΩ
GND

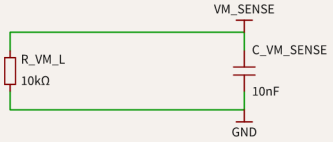

U_DRV
VCP
C_CP
100nF
VREG
B_MINUS
ENABLE_N
CP1
CP2
OUT2B
ENABLE
GND2
CP1
CP2
VCP
NC3
VREG
MS1
MS2
MS3
RESET
ROSC
SLEEP
VDD
PAD
VBB2
SENSE2
OUT2A
NC1
OUT1A
SENSE1
VBB1
OUT1B
NC2
DIR
GND1
REF
STEP
V3V3
GND
SLEEP
VREF
GND
A4988SETTR-T

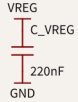

J_MOTOR
A_PLUS
A_MINUS
B_MINUS
B_PLUS
B4B-PH-K-S(LF)(SN)

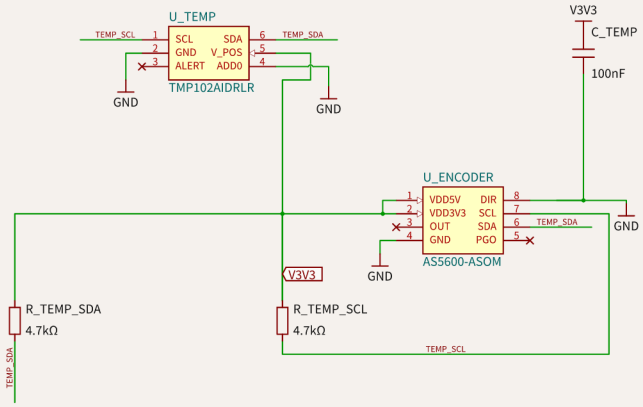

SENSE1
R_SA
240mΩ
SENSE2
R_SB
240mΩ
GND

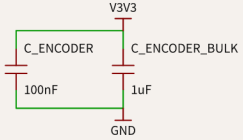

VCP
C_VCP
100nF
PD_VBUS

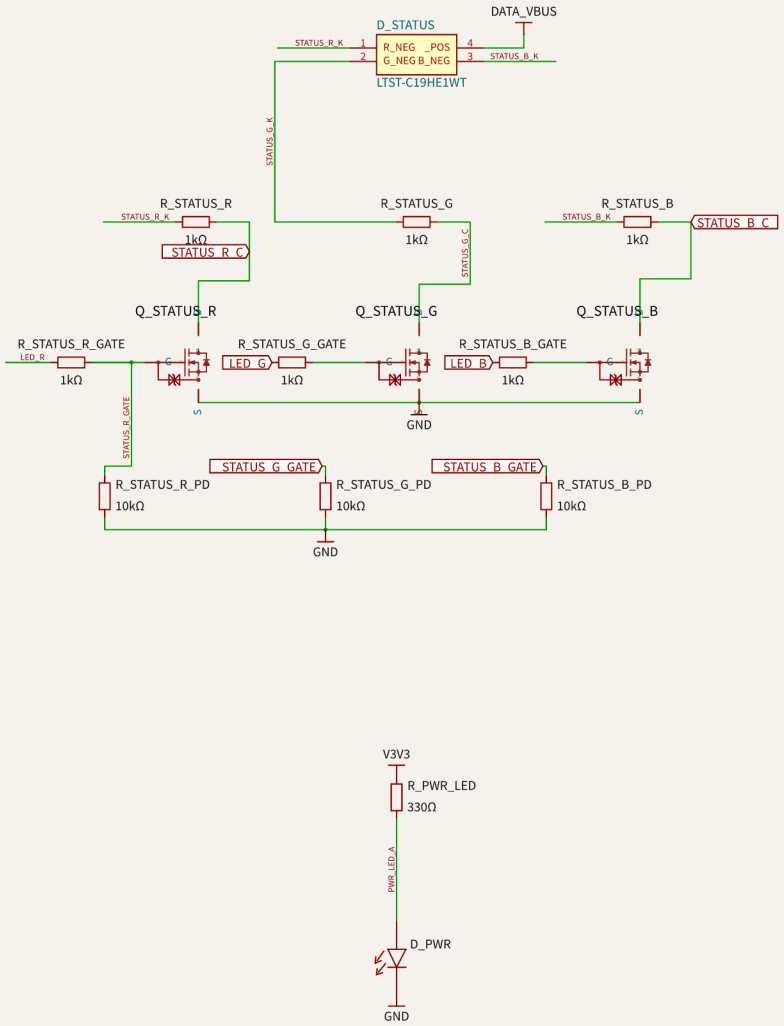

PD_VBUS
VM_SENSE
R_VM_H
100kΩ

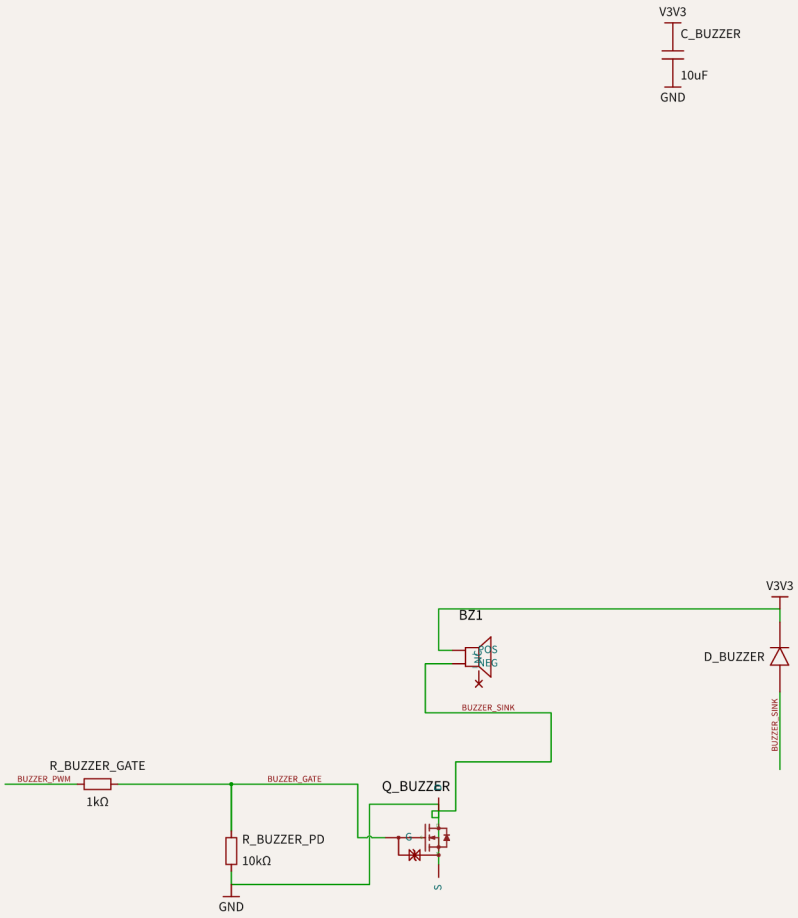

VM_SENSE
R_VM_L
10kΩ
C_VM_SENSE
10nF
GND


VREG
C_VREG
220nF
GND


U_TEMP
TEMP_SCL
SCL
GND
ALERT
SDA
V_POS
ADD0
TEMP_SDA
TMP102AIDRLR
GND
GND
V3V3
C_TEMP
100nF
U_ENCODER
VDD5V
VDD3V3
OUT
GND
DIR
SCL
SDA
PGO
TEMP_SDA
AS5600-ASOM
GND
GND
V3V3
R_TEMP_SDA
4.7kΩ
R_TEMP_SCL
4.7kΩ
TEMP_SDA
TEMP_SCL


V3V3
C_ENCODER
100nF
C_ENCODER_BULK
1uF
GND


D_STATUS
DATA_VBUS
STATUS_R_K
R_NEG _POS
G_NEG B_NEG
STATUS_B_K
LTST-C19HE1WT
STATUS_G_K
R_STATUS_R
R_STATUS_G
R_STATUS_B
1kΩ
STATUS_R_C
STATUS_G_C
STATUS_B_C
Q_STATUS_R
Q_STATUS_G
Q_STATUS_B
R_STATUS_R_GATE
R_STATUS_G_GATE
R_STATUS_B_GATE
LED_R
LED_G
LED_B
GND
STATUS_R_GATE
STATUS_G_GATE
STATUS_B_GATE
R_STATUS_R_PD
R_STATUS_G_PD
R_STATUS_B_PD
10kΩ
V3V3
R_PWR_LED
330Ω
PWR_LED_A
D_PWR


V3V3
C_BUZZER
10uF
GND
V3V3
BZ1
D_BUZZER
BUZZER_SINK
BUZZER_SINK
R_BUZZER_GATE
BUZZER_PWM
1kΩ
BUZZER_GATE
Q_BUZZER
R_BUZZER_PD
10kΩ
GND